

Test plan ID and title: BMI Test Plan V1

James Heinrichs

Alana Heinrichs

1. Introduction

The test plan is to test various situations in utilizing a BMI application. The intent is to confirm functionality of the application and solution for both valid and invalid inputs.

2. Scope of Testing

- **In Scope:** Unit testing for the BMI calculation formula (weight/height²), boundary value testing for category classifications (underweight, normal, overweight, obese), and input validation (handling strings, negative numbers, or zeros).
- **Out of Scope:** Performance load testing and cross-platform GUI rendering (if building a command-line interface).

3. Test objectives and approach: Automated testing with PYTEST to execute the scripts.

4. Test schedule and milestones: Two days before the due date for the project the test cases will be implemented.

5. Test environment setup: PC utilizing a PYTHON Compiler, GITHUB repository

6. Resources and responsibilities: James Heinrichs, lead software developer overseeing results defects, Alana Heinrichs Lead tester documenting findings.

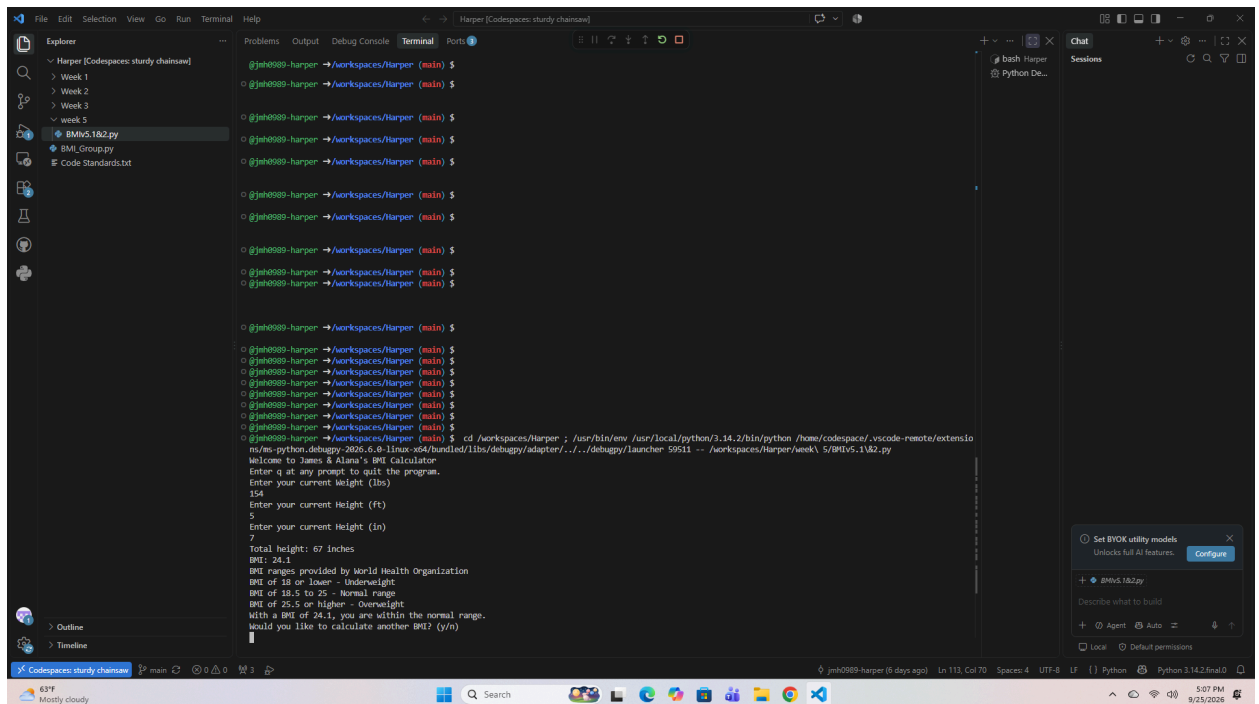
7. Test deliverables: Screen images and versioning stored in GITHUB repository with version tracking.

8. Entry and exit criteria: Specifies the conditions required to begin testing and the conditions that must be met to complete it.

9. Risks and mitigation strategies: Identifies potential blockers, constraints, or failure points and how the team plans to address them.

Test Cases

- **Valid Normal Weight (Imperial):** Input weight = 154 LBS, height = 5ft 7in. Expected BMI = 24.1, Category = Normal Weight. [1]



- **Valid Underweight (Boundary):** Input weight = 90 LBS, height = 5ft 0in . Expected BMI = 17.6, Category = Underweight. [1]

- **Zero or Negative Input:** Input height = 0 or -1.75. Expected outcome = Error caught or prompt to re-enter valid data without crashing. [1]

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@joh0989-harper → /workspaces/Harper (main) $ cd /workspaces/Harper ; /usr/bin/env /usr/local/python/3.14.2/bin/python /home/codespace/.vscode-remote/extensions/ms-python.debugpy-2026.6.0-linux-x64/handled/lib/debugpy/adapter/.../debugpy/launcher 59511 -- /workspaces/Harper/week5/BMIv5.1&2.py
Enter your current Height (ft)
5
Enter your current Height (in)
7
Total height: 67 inches
BMI: 24.1
BMI ranges provided by World Health Organization
BMI of 18 or lower - Underweight
BMI of 18.5 to 25 - Normal range
BMI of 25.5 or higher - Overweight
With a BMI of 24.1, you are within the normal range.
Would you like to calculate another BMI? (y/n)
y
Enter your current Weight (lbs)
90
Enter your current Height (ft)
5
Enter your current Height (in)
6
Total height: 60 inches
BMI: 17.6
BMI ranges provided by World Health Organization
BMI of 18 or lower - Underweight
BMI of 18.5 to 25 - Normal range
BMI of 25.5 or higher - Overweight
With a BMI of 17.6, you are Underweight.
Would you like to calculate another BMI? (y/n)
y
Enter your current Weight (lbs)
176
Enter your current Height (ft)
5
Enter your current Height (in)
7
Total height: 67 inches
BMI: 27.6
BMI ranges provided by World Health Organization
BMI of 18 or lower - Underweight
BMI of 18.5 to 25 - Normal range
BMI of 25.5 or higher - Overweight
With a BMI of 27.6, you are Overweight.
Would you like to calculate another BMI? (y/n)
y
Enter your current Weight (lbs)
176
Enter your current Height (ft)
0
Please enter a number greater than 0, or enter q to quit.
Enter your current Height (ft)
1

```

- **Non-Numeric String Input:** Input weight = abc. Expected outcome = ValueError caught and handled gracefully.

